In [1]:
```
number1=eval(input("enter number1:"))  # number1='100'
number2=eval(input("enter number2:"))  # number2='200'
print(f"the addition of {number1} and {number2} is {number1+number2}")
```

```
enter number1:20
enter number2:30
the addition of 20 and 30 is 50
```

- Syntax error

```
    - if you miss any :,),""
```

- Name error
  - Variable is not defined in above lines
- Type error
  - when we convert one type to anothr type
- Value error
  - when you do maths , you might not do proper math operations
- Attribute error
  - the package does not has method name
- Zero division error
  - try to divide by zero
- Indentation error
  - space
  - is a kind of syntax error

In [ ]:
```
assume that you write 1000 lines of code
suppose at 500 line you got error
but you belive that it is a not a generic error,
if at that point error is okay becuase other codes are not affect

you want to run all 1000 lines of codes even though you got error
at one point
```

In [ ]:
```
try part
==== original code will write===========
except part
====== error will write =========
```

In [6]:
```
number1=eval(input("enter number1:"))  # 10
number2=eval(input("enter number2:"))  # python
print(f"the addition of {number1} and {number2} is {number1+number2}")
```

```
enter number1:10
enter number2:python

---------------------------------------------------------------------------
-
NameError                                 Traceback (most recent call las
t)
Cell In[6], line 2
      1 number1=eval(input("enter number1:"))
----> 2 number2=eval(input("enter number2:"))
      3 print(f"the addition of {number1} and {number2} is {number1+number
2}")

File <string>:1

NameError: name 'python' is not defined
```

In [9]:
```
try:
    number1=eval(input("enter number1:"))
    number2=eval(input("enter number2:"))
    print(f"the addition of {number1} and {number2} is {number1+number2}")

except:
    print("Name error")
    print("hello")
    n1=10
    n2=20
    print(n1+n2)
```

```
enter number1:10
enter number2:python
Name error
hello
30
```

In [19]:
```
try:
    number1=eval(input("enter number1:"))
    number2=eval(input("enter number2:"))
    print(f"the addition of {number1} and {number2} is {number1/number2}")

except Exception as e:
    print(e)
    ########
    ########
```

```
enter number1:10
enter number2:p
name 'p' is not defined
```

- you can write any lines of code in try block and except block
- but python is a step by step process
- first it will try to execute try block
- if you get any error in try block , then immediately it will got to except block

- till the error in try block , that lines of codes execute
- whenwever error ocured, excepect block codes will execute
- it will not handle syntax errors

In [16]:
```
num1=10
num2=20
print("the addition of {} and {} is {}".format(num1,num2,num1+num2))
```

```
the addition of 10 and 20 is 30
```

In [ ]:
```
1. Can we use “try block” without a “except block”
2. And “catch block” without “try block”
```

In [17]:
```
try:
    number1=eval(input("enter number1:"))
    number2=eval(input("enter number2:"))
    print(f"the addition of {number1} and {number2} is {number1/number2}")
```

```
  Cell In[17], line 4
    print(f"the addition of {number1} and {number2} is {number1/number2}")

^
SyntaxError: incomplete input
```

In [21]:
```
try:
    name=input("enter your name:")
    age=int(input("enter your age:"))
    city=input("enter your city:")
    salary=float(input("enter your salary:"))
    print(f"My name is {name}, I'm {age} years old, I came from {city}, I e

except Exception as e:
    print(e)
```

```
enter your name:p
enter your age:thirty
invalid literal for int() with base 10: 'thirty'
```

In [22]:
```
try:

    name=input("enter a name:")
    if name=="python":
        print("hai")
        print(1)
        print("good")
    else:
        print("enter a valid name")

except Exception as e:
    print(e)
```

```
enter a name:python
hai
1
good
```

In [24]:
```python
try:
    num=eval(input("enter a number"))
    if num%2==0:
        print("it is an even")
    else:
        print("it is an odd number")

except Exception as e:
    print(e)
```

```
enter a number10
it is an even
```

In [ ]:
```python
import random

try:
    num=random.randint(1,50)
    if num%2==0:
        print(f"{num} is even")
    else:
        print(f"{num} is odd")

except Exception as e:
    print(e)
```

In [ ]:
```python
try:
    name = input("enter a name: ")
    salary = eval(input("enter a salary: "))
    tax_percentage = eval(input("enter a tax in %: "))
    tax_pay = (salary * tax_percentage)/100

    if tax_pay >= 100000:
        print(f"{name} earns good")
    else:
        print(f"{name} earns less")

except Exception as e:
    print(e)
```

In [ ]:
```python
try:
    num=random.randint(1,200);
    if num>99:
        print(f"{num} is greater then 99")
    else:
        print(f"{num} is less then 99")

except Exception as e:
    print(e)
```

In [26]:
```
try:
    num=eval(input("enter the number:"))
    if num>0:
        print(f"{num} is a postive number")
    elif num<0:
        print(f"{num} is a negative number")
    else:
        print("it is zero")

except Exception as e:
    print(e)
```

```
enter the number:p
name 'p' is not defined
```

In [ ]: